

A Dataset and Machine Learning Approach to Classify and Augment Interface Elements of Household Appliances to Support People with Visual Impairment

Hanna Tschakert
LMU Munich
Munich, Germany
hanna.tschakert@web.de

Florian Lang
LMU Munich
Munich, Germany
florian.lang@ifi.lmu.de

Markus Wieland
VISUS, University of Stuttgart
Stuttgart, Germany
markus.wieland@visus.uni-stuttgart.de

Albrecht Schmidt
LMU Munich
Munich, Germany
albrecht.schmidt@ifi.lmu.de

Tonja-Katrin Machulla
TU Chemnitz
Chemnitz, Germany
tonja.machulla@phil.tu-chemnitz.de

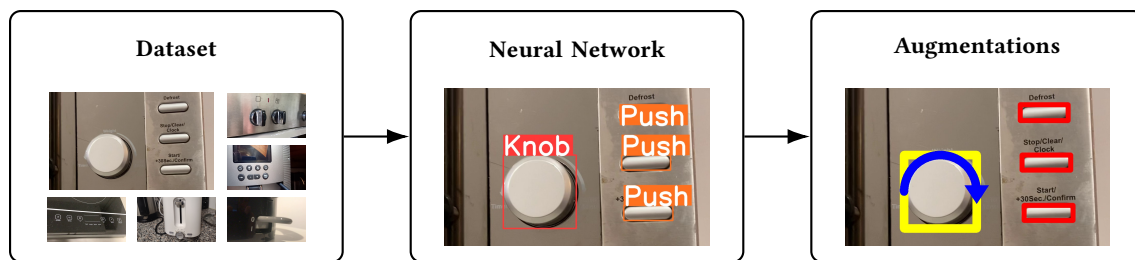


Figure 1: Here, we show the three main steps of our work. First, we created a dataset with 13,702 images of interfaces from household appliances and annotated the interface elements. All images and additional data are available on Github [39, 40]. A selection of images is shown in the left box. The middle box shows the output of a neural network that we trained to recognize *PushButtons*, *Knobs*, *TouchButtons*, *Sliders*, and *Toggles*. Finally, we designed high-contrast AR augmentations to augment the interface elements to support users with visual impairment. Such an augmented image is shown in the right box.

ABSTRACT

Many modern household appliances are challenging to operate for people with visual impairment. Low-contrast designs and insufficient tactile feedback make it difficult to distinguish interface elements and to recognize their function. Augmented reality (AR) can be used to visually highlight such elements and provide assistance to people with residual vision. To realize this goal, we (1) created a dataset consisting of 13,702 images of interfaces from household appliances and manually labeled control elements; (2) trained a neural network to recognize control elements and to distinguish between *PushButton*, *TouchButton*, *Knob*, *Slider*, and *Toggle*; and (3) designed various contrast-rich and visually simple AR augmentations for these elements. The results were implemented as a screen-based assistive AR application, which we tested in a user study with six individuals with visual impairment. Participants were

able to recognize control elements that were imperceptible without the assistive application. The approach was well received, especially for the potential of familiarizing oneself with novel devices. The automatic parsing and augmentation of interfaces provide an important step toward the independent interaction of people with visual impairments with their everyday environment.

CCS CONCEPTS

• **Human-centered computing** → **Accessibility technologies; Accessibility systems and tools.**

KEYWORDS

Dataset, Neural Network, Object Recognition, Household Appliance, Visual Impairment, Augmented Reality, Low Vision, Accessibility

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1 INTRODUCTION

Interacting with new household appliances can be bothersome due to their complex functionality and non-intuitive labeling of interface elements. For the over 270 million [1] individuals with severe visual impairment or blindness, this task becomes extremely challenging. Due to the continuous increase in the number of people with visual impairments [32] and the low-contrast design and touch elements in modern devices, this issue becomes increasingly important. People with visual impairment struggle with the intuitive usage of such devices [26, 27]. Interface elements that lack tactile feedback, such as touch elements, or elements that combine different types of interactions, such as pressing and turning, prove challenging. Henceforth, people with visual impairments must use assistive devices or ask others for help when familiarizing themselves with new appliances.

This hurdle can be lowered by integrating the appliances into smart home systems allowing for other input modalities, for example, smartphones or gestures [2, 5, 6, 19, 38]. Another option is to label the appliances with tags created manually [9, 33] or using neural networks [35, 45], which then offer explanations and support to the user.

Most of these systems require a customization of the appliances or a connection to the home network. However, unfamiliar interfaces cause the most difficulties for people with visual impairment and benefit the most from assistance [23]. In contrast to their own appliances, users have no opportunity to adjust or learn the interface of unfamiliar appliances encountered in other households, at work, or in public. So while an assistive device should provide support for known appliances and confirm the user's actions, it should focus on unfamiliar appliances [25].

We envision an augmented reality (AR) application that can identify the interaction type and function of interface elements as well as the current state of the household appliances and provides guidance to the user. Approaches combining manual labeling with computer vision have been demonstrated by Guo et al. They use crowd-sourcing to label interfaces and machine learning to recognize previously labeled interfaces [10, 11]. However, we envision a system that combines object recognition, text recognition, and machine learning to remove the need for human input.

In our work, we lay the foundation for the creation of such a system. First, we created a dataset containing 13,702 images of interfaces from household appliances and labeled all interface elements with their size and position in the image as well as their interaction type. Here, we divided elements into five types, namely *PushButton*, *TouchButton*, *Knob*, *Slider*, and *Toggle*. Then, we trained a YOLOv5s neural network [42] in 150 epochs to find and classify interface elements using our dataset. Utilizing this neural network, we implemented a screen-based AR application that finds and highlights interface elements in a video stream. As many people with visual impairment have residual vision and actually prefer using their visual sense wherever possible [25], we choose to implement contrast-rich visual feedback. For that purpose, we created a design space for visual augmentations of the interface elements. Based on the design space, we implemented 18 visualizations in a prototype application to indicate the location and the type of interaction for the different interface elements. The dataset, the

neural network model, and the software are published on GitHub¹ [40] and mirrored on Zenodo² [39].

We then conducted a user study with six individuals with visual impairment. We collected feedback regarding the different visualizations and real-world helpfulness of our approach. Our results show that our visual augmentations are recognizable and indicate the interaction type. As part of the study, participants were asked to explore a real kitchen. Here, they could perceive previously imperceptible interface elements and their interaction type using our prototype.

Our contribution presented in this work is threefold: (1) a labeled dataset for interfaces of household appliances (2) a trained neural network and a prototype application showcasing the use of our dataset, and (3) a user study with six participants with visual impairments showing that the automatic parsing and augmentation of interfaces can support their interaction with their everyday environment.

2 RELATED WORK

We base our work on research in the field of assistive AR and assistive technology for operating household appliances. However, research in other fields also explores the combination of AR and objects in the real world to create new interfaces. For example, Henderson and Feiner [17] used objects in the environment to provide tactile feedback for a virtual interface, and Hettiarachchi and Widgor [18] mapped virtual objects onto real-world objects with a similar shape, to allow for tangible manipulation of the virtual objects.

Assistive Augmented Reality. Assistive AR applications are frequently used in other work [8, 23, 28, 36, 47, 48]. They utilize the remaining vision of a visually impaired user by augmenting the real world with additional or otherwise hard-to-see information about the environment. Providing such visual feedback requires less cognitive load than auditory feedback once the meanings of different visualizations have been learned [45]. Further, AR helps to reduce the cognitive effort when operating touchscreens [23] and increases the sense of safety and confidence when climbing stairs [47]. When providing visual feedback, interindividual differences in visual perception and personal preferences lead to differences in the preferred augmentations [23, 48]. Therefore, the user should be allowed to customize the augmentations to their individual needs [14]. However, there is a common tendency not to obscure the augmented object [23, 36], as well as to avoid movement in the visualization [23, 48]. In the study of [30], they combine an AR device with an object recognition feature. The AR application provides simple visual feedback in the form of a red rectangle along with audio feedback. In a user study, nine out of 15 people with visual impairment indicated that the device was helpful. Even though the authors used the same visually simple visual representation for all recognized objects, the results show that such an AR device is perceived positively by people with visual impairments. Gurari et al. [13] proposed a different application where people with visual impairments can upload images of their environment and ask questions, e.g., if they cannot recognize an object. Using

¹GitHub: <https://github.com/FlorianLa/dataset-interfaces-household-appliances> [40].

²Zenodo: <https://doi.org/10.5281/zenodo.7586106> [39].

this application, they created a dataset of over 30,000 images with a spoken question and 10 crowd-sourced answers. They continued by labeling these images regarding privacy issues in a more recent publication [12].

Assistive Technology for Operating Household Appliances. Assistive devices to support people with visual impairments in operating household appliances are often integrated into smart home systems. They circumvent the appliance’s interface and offer the user control through their mobile devices [5, 19, 38] or finger gestures [2]. Alternatively, devices can be tagged with information on how to interact with the device. These tags can be generated manually [33] or with the help of a crowd-based system [9]. One can also analyze the energy consumption to identify appliances and their current state using the Bayes Classification Algorithm [44] or the neural network WaveNet [20]. Neural networks can also distinguish body poses and then infer the household appliance the user is currently operating [45] or recognize the type of appliance [37]. After information about the household appliance is collected, one can inform users about the type of device and provide general information about its usage [9, 33, 37]. Alternatively, the user can be guided towards the relevant interface element either using audio [10, 11] or visual [23] feedback.

3 DATASET

Training a neural network requires a large amount of data. While there are large datasets to learn object recognition, like ImageNet [22], often authors have to create their own dataset [13, 37, 45] for more specific use cases. To our knowledge, there is no collection of images focusing on interface elements of household appliances. Further, the image quality of ImageNet was insufficient to extract the interfaces from the images containing household appliances. Therefore, we collected and annotated 13,702 images of interfaces from household appliances.

To allow for publishing the dataset, we only used images where we hold the copyright. To that extent, we asked 40 friends and colleagues to submit images from household appliances that not only contain a touchscreen interface, i.e., contain at least one fixed control element. Further, conditions in the real world are never optimal. The user might look at the device from the side and lighting conditions vary. Therefore, we instructed the contributors to film the interface while moving in a circle around it and create multiple videos with different lighting conditions (daylight, dusk, artificial light). In total, we collected 331 videos and three photo sequences of 245 different household appliances and 32 different types of household appliances. Then, we extracted around 30 to 55 images from each video, showing the interface from different angles, see Figure 2. Individual frames in which we could not identify interface elements or where the person filming was visible were discarded, resulting in a total of 13,702 images.

During the next step, we labeled the images manually using Labelling [41]. For each interface element, we defined a rectangular bounding box that completely encloses the element. Based on the collected images, we identified four types of interaction; turning, sliding, flipping, and pushing. Accordingly, we annotated each interface element with one or more of the following classes: *Knob*, *Slider*, *Toggle*, *PuchButton*, and *TouchButton*. Here, we split the class

of *Buttons* due to the large visual discrepancy between actual tactile buttons, which need to be pushed in and touch elements on a flat surface.

The dataset contains 75,551 labeled interface elements. In Table 1, we list the exact distribution of the five different classes of interface elements and the lighting conditions under which the images were taken. The images are saved in JPG format, and the label information is provided in TXT, JSON, and XML format. The full dataset is published on GitHub [39, 40] so that other researchers can use and extend the collection.

4 NEURAL NETWORK

Neural networks are mathematical functions that receive an input, i.e., an image, and produce an output, i.e., a list of interface elements. Neural networks consist of so-called nodes, which are connected via edges and can receive, modify, and send information from and to other nodes [43]. Each edge is assigned a specific weight which influences the transfer of information between nodes. Training a network refers to adjusting the weights of all edges. In the training phase, the network is provided test data, i.e., pairs of input and desired output. With that data, the weights of the network are optimized so that the neural network produces the desired output from a given input. To reduce an array of pixels to the desired output, Convolutional Neural Networks are often used [3]. Such a network has multiple layers, each consisting of differently structured nodes, and information is passed from layer to layer.

In our case, we trained a neural network to identify the position, size, and interaction type of interface elements in images. While we implement our prototype on a notebook computer, future systems are supposed to run on mobile devices or head-mounted displays. Therefore, we selected Yolov5s, the second smallest model of Yolov5 [42]. With a low computational load in mind, we choose a *depth_multiple* (number of layers) of 0.33 and a *width_multiple* (number of nodes) of 0.50. Yolov5s consists of six different building blocks, specifically, a standard 2D convolution layer (Conv) [4], a Cross Stage Partial bottleneck with three Convs (C3), a Spatial Pyramid Pooling - Fast layer which was implemented for Yolov5 (SPPF), the Nearest Neighbor algorithm for upsampling (Upsample), a concatenation of tensor lists along the dimension (Concat), and an application of Convs on the inputs before collecting all outputs (Detect). The structure of the neural network is shown in Figure 3.

4.1 Training

For training our network, we first divided the images into three subsets. We used about 70% of the images for training, 20% for validation, and 10% as test set. The training set includes 9,598 images, the validation set 2,748 images, and the test set contains 1,356 images. However, as images taken from the same video are highly similar, we divided the images based on their source video, i.e., all images generated from one video are in the same subset, which resulted in an uneven distribution for some classes. We list the detailed distribution of interface elements over these three sets in Table 2.

When training a neural network, we have to specify a learning rate, which determines how fast the weights change during each training step. To reduce instabilities in early training [42], prevent



Figure 2: Here we show four images of a stove's interface. The three left images were extracted from one video recorded under daylight and the right image is extracted from a different video of the same device, recorded under artificial light. All images show the interface from different angles.

Table 1: Here, we list the distribution of the number of the five different control element types *Knob*, *Slider*, *Toggle*, *PushButton*, and *TouchButton* in the dataset. We separated them into the three different lighting conditions.

	<i>knob</i>	<i>Slider</i>	<i>Toggle</i>	<i>PushButton</i>	<i>TouchBotton</i>	Σ
Daylight	8,528	774	805	23,177	3,363	36,647
Artificial Light	6,536	293	879	21,590	3,348	32,646
Bad Light	2,693	36	75	3,442	12	6,258
Σ	17,757	1,103	1,759	48,209	6,723	75,551

Table 2: This Table shows the distribution of images and the five different interface element types, namely *PushButton*, *Knob*, *TouchButton*, *Slider*, and *Toggle*, into the training, validation, and testing set. The last column lists the total number of interface elements contained in the respective set.

	Images	<i>PushButton</i>	<i>Touchbutton</i>	<i>Knob</i>	<i>Slider</i>	<i>Toggle</i>	Σ
Training	9,598	37,194	1,560	15,238	747	1,363	56,102
Validation	2,748	6,841	3,077	1,878	155	310	12,261
Testing	1,356	4,174	2,086	641	201	86	7,188

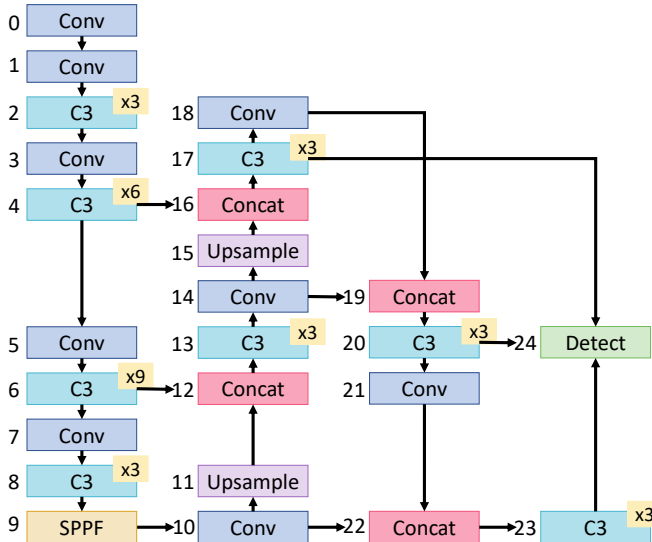


Figure 3: Here, we show the structure of the YOLOv5s model used for our application. The model consists of six different building blocks coded in different colors.

early overfitting, and achieve a better generalization instead of pure memorizing [15, 29], we set the learning rate to a high value of 0.1 for the first three training epochs. Afterward, we lowered it to 0.01. We chose a batch size of 32, i.e., weights are updated after every 32 images processed. As the mathematical algorithm to optimize the weights during training, we used a Stochastic Gradient Descent (SGD) since it achieves a shorter training time than typical gradient descent algorithms [16].

In total, we used 150 training epochs. We started the first iteration with a pre-trained model provided by the developers of YOLOv5 [42]. We investigated the influence of further training and determined that the model did not improve after an additional 120 epochs. Therefore, we continued with the best model after 150 training epochs.

4.2 Technical Evaluation

While the training and validation set are used during the training of the neural network, the test dataset contains images that the algorithm has not previously seen. We used the test dataset to evaluate our model.

The images in our dataset often contain *Knobs* and they are usually similarly shaped, leading to the best recognition rate out of all interface elements (0.88 on the test set). The most common



(a) Here, four images from the dataset showing different toasters are displayed. Each interface contains a *Slider*. However, the sliders differ significantly in design.



(b) Again, we show four images from the dataset. All images show a kettle, and the interfaces contain one *toggle* each. The *Toggles* are all visually similar.



(c) Here, we display two examples of *Toggles*, which are visually more similar to *PushButtons* rather than to the majority of *Toggles*. Such designs lead to wrong classifications.



(d) This image shows an interface element from the dataset. The outer ring can be turned and used like a *Knob*, while the center functions as a *PushButton*.

Figure 4: The four parts of this figure show examples from the dataset, which might have influenced the detection of interface elements by the neural network. Part (a) on the top left shows the varying design of *Sliders*, (b) on the top right displays the similarity of most *Toggles*, and (c) on the bottom left shows the *Toggles* that are visual outliers. Lastly, part (d) on the bottom right shows an example of an interface element belonging to two classes.

elements in our dataset are *Buttons*. While they are generally recognized well, *TouchButtons* are visually different and often blend into the background. They are frequently classified as *PushButton* and generally recognized with less accuracy. The dataset contains fewer *Sliders*, and they are often visually different in form and size, see Figure 4a. This leads to a bad recognition rate for *Sliders* (0.08 on the test set). There are also few *Toggles* contained in the dataset. However, they often belong to the same type of household appliance, a kettle, and are visually similar, see Figure 4b, leading to a slightly better detection rate (0.31 on the test set). Only *Toggles* with high similarity to a *PushButton* were classified wrongly on a regular basis (0.20 on the test set), see Figure 4c. We show the confusion matrices after Kaanakan [21] and the detailed accuracy values regarding the test and validation set in Figure 5.

There were also some difficulties in the recognition of atypical buttons. For example, some interface elements can be used as a *Knob* and turned, but also pushed and used as a *PushButton* as shown in Figure 4d. As we did not account for elements in multiple classes, the classification switched back and forth between *Knob* and *PushButton* depending on the camera angle and the distance to the element. Also, atypical shapes, especially for *PushButton*, caused difficulties, and the model did not recognize them correctly. Further, the distribution and number of the different classes in the training, validation, and test set mentioned in the last section may have impacted the recognition of *Sliders*, *TouchButtons*, and *Toggles*.

Despite these difficulties, our system was able to recognize all types of control elements used during the study with sufficient accuracy. The system works most optimally between a five and 50cm distance between the camera and interface and under good lighting conditions. The recognition rates can be improved by using more training data or by more elaborate post-processing of the detected elements.

4.3 Prototype Application

In the next step, we implemented a prototypical application utilizing our model. We used a handheld camera connected via USB to a computer to film the interface elements of household appliances. We chose this setup over a prototype using augmented reality glasses, as it allowed for more rapid prototyping, resembles the first part of the study more closely, and has fewer constraints regarding computational resources.

The user can select and customize visualizations using different parameters for each class. We choose to combine *TouchButton* and *PushButton* into *Button* and augment them in the same manner. The computer then displays the live feed of the camera image with a delay of 170ms to 500ms. For each video frame, the system locates and identifies interface elements and visually highlights them. Here, we choose to highlight only interface elements where the neural network returns a probability of at least 0.6, and the resulting bounding box is not too small.

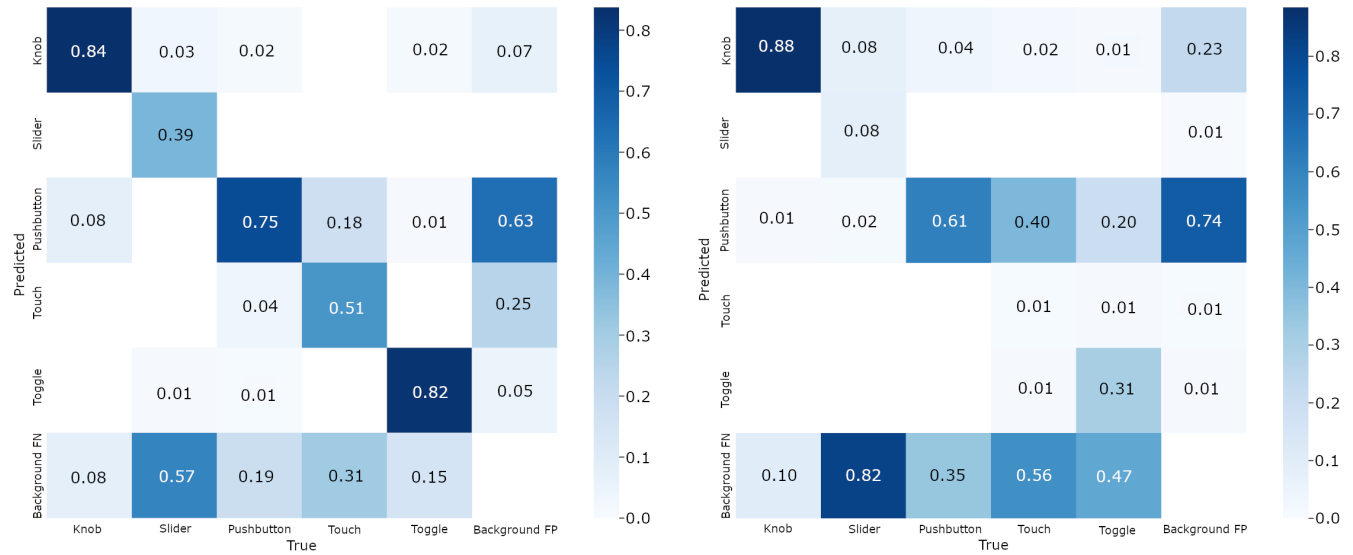


Figure 5: Here, we show the confusion matrices calculated after Kaanakan [21]. The left matrix is calculated based on the validation set and the right matrix on the (unseen) test set. We only included interface elements in this calculation, where the model showed a certainty of at least 25%. Further, the ratio between the area of overlap between a recognized bounding box and a bounding box from the ground truth to the combined area (intersection of union, IoU) had to be at least 0.5. The matrices show, that *Sliders* and *Toggles* are recognized less compared to *Knobs* and *Buttons*.

5 DESIGN SPACE AND AUGMENTATIONS

We created a design space for visual augmentations of the different interface elements. Here, we derived the key aspects of such visual designs from related work. For example, we restricted the design to static augmentations, as motion can be perceived as distracting [23, 48]. Afterward, we implemented 18 concrete augmentations, which cover most aspects of the design space to facilitate a broad evaluation.

5.1 Design Space

In our work, we want to design augmentations that are displayed simultaneously for all elements of an interface. Therefore, we limit the augmentation to the size of the bounding box of each element. Since color and brightness are crucial for the visual perception of people with visual impairments [25], we centered our design space around these parameters. The form of an object only plays a subsidiary role in visual perception [25]. Therefore, we chose a rectangular shape, following the detected bounding box. The border of this rectangle can be drawn solid, dashed, or dotted in various different thicknesses. However, depending on the device’s color, there could be little contrast between the device and the border. To mitigate this issue, we added a filling to the augmentation. These solid-colored rectangles can create a strong contrast to the border and can support the differentiation between elements on the interface. To further support the differentiation between different types of elements and to indicate their usage, we added arrows to *Toggles*, *Sliders*, and *Knobs*. In the latter case, we also included curved arrows to reflect the rotational movement. We also included the option to

add a label to leverage the benefits of a textual representation of information compared to symbolical representation [24].

All aspects, such as the border, filling, arrow, and arrowhead can be independently adjusted in brightness and color. The filling of the augmentation can be adjusted in transparency and size, i.e., if it covers the whole box or only the central area. When combining the border with a filling, the same, different, or complementary colors can be chosen. Lastly, the arrow can be straight or curved with different shapes of arrowheads. The color and thickness of the arrowhead can be chosen independently of the arrow line. The arrow can also be positioned on the border of the augmentation or at some distance. The entire design space is shown in Table 3.

5.2 Selected Augmentations

To evaluate our design space, we created 18 different augmentations. Based on the number of occurrences in household appliances, seven augmentations were created for *Knobs* and *Buttons* and two for *Toggles* and *Sliders*. We designed these augmentations so that they cover most aspects of the design space in different forms and complement each other. We also drew inspiration and necessary requirements, i.e., high contrast, from related works [23, 34, 36, 47, 48]. We show all 18 augmentations in Figure 6.

6 USER STUDY

The design of our augmentations and our prototype application leads to three research questions that we want to answer by conducting a user study with people with visual impairment. **RQ1:** Are the augmentations recognizable for people with low vision? **RQ2:** Are the augmentations helpful and provide an advantage

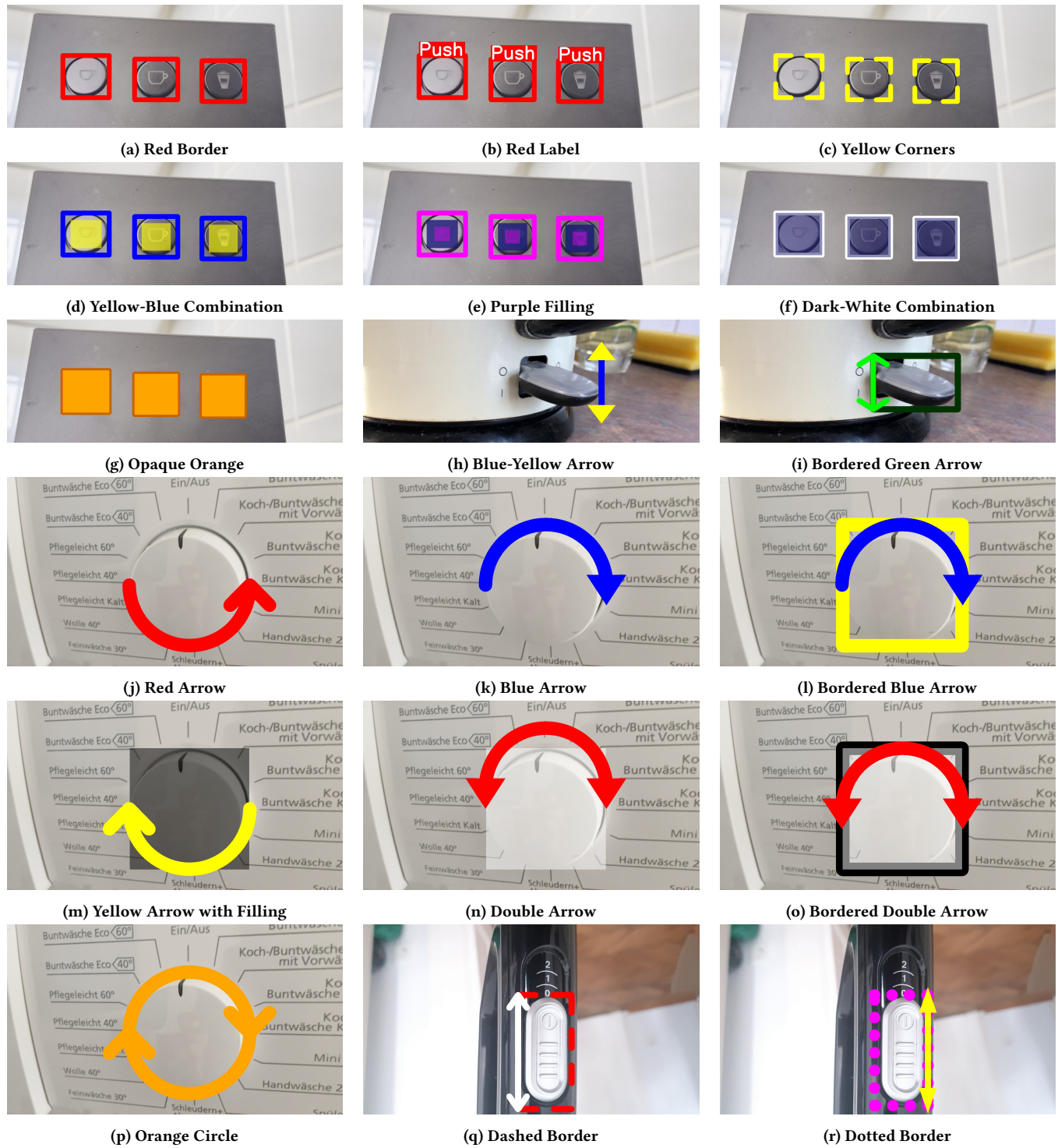
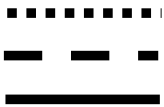
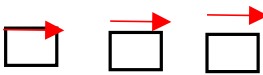
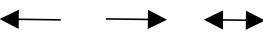


Figure 6: This figure shows all 18 augmentations implemented and used during the user study. From left to right and top to bottom, we first see seven augmentations for *Buttons*, two augmentations for *Toggles*, seven for *Knobs*, and two for *Sliders*. The augmentations were created based on our design space and differ in their border, filling, and arrow. We chose and combined these aspects so that they cover most possible parameter values of our design space.

Table 3: This table shows our design space. The left column describes the overall aspect of an augmentation and the second column the concrete design parameter. The third column lists the possible range of values for that parameter, and the last column visualizes an example of the possible parameter values.

	Parameter	Possible Values	Example
General	Color	White – Color – Black	
	Brightness	Light – Dark	
Border	Line type	Dotted Dashed Solid	
	Thickness	Thin – Thick	
	Label	Without ; Text	Push
Filling	Size	Small area in the middle – entire inner field	
	Color depending on the border	Same – Complementary	
	Transparency	Transparent – Opaque	
Arrow	Arrow head, shape	Strokes – Triangles	
	Thickness	Thin – Thick	
	Arrow head, color	Same as line – Complementary to line	
	Distance to the border	Near – Far	
	Orientation	Left – Right – Double-sided	
	Curvature	Curved ; Straight	

when interacting with unfamiliar devices? **RQ3:** Are the different augmentations sufficient to distinguish between the different types of interface elements?

To answer these questions, we structured our user study into three sections. First, we showed images of interface elements to the participant. For each element, all applicable augmentations were shown after each other. Secondly, we led the participant into a real kitchen and let them explore the kitchen with our prototype application using their preferred augmentation. Lastly, we showed images of more complex interfaces containing multiple types of elements to the participant.

6.1 Participants

We recruited six participants with visual impairments by contacting local support groups for people with visual impairments. We also invited participants from previous studies who expressed their

interest in future studies. Four participants identified as female, two as male, and none as diverse. The mean age of the participants was 48.3 years, with a standard deviation of 30.4 years. All participants were financially compensated for their participation with 12 EUR per hour. The participants had different diagnoses, and their visual acuity ranged from 2% to 20%, see Table 4. Participant P6 was asked to cover one eye during the study, as he still has 60% to 70% visual acuity in his left eye. The study was approved by the Institutional Review Board of the authors' department.

6.2 Technical Setup

For the first and third sections of the study, the user was seated in front of an iiyama ProLite full HD touch display with a screen diagonal of 21.5 inches and a resolution of 1920x1080 pixels. The displayed JPG images were selected from our dataset and augmented

Table 4: Here, we list the detailed demographic and medical data for all participants. If multiple medical conditions were reported, we list the one with the greatest impact on the participant’s sight. AMD is the abbreviation for age-related macula degeneration. For P4 and P6 we list the left (l) and right (r) eyes separately.

ID	Age	Gender	Visual Acuity	Diagnosis
P1	22	m	0.03	Liver congenital amaurosis
P2	51	w	0.03	Cataract and glaucoma
P3	24	w	0.12	Achromatopsia
P4	85	w	l: 0.07; r: 0.03	Dry AMD
P5	23	w	0.02	Cone-rod dystrophy
P6	85	m	l: 0.65; r: 0.20	l: dry AMD; r: wet AMD,

with pre-computed visualizations. For the second part, the participant walked around in the kitchen of our department. Here, we used an Acer notebook with an x64-based processor and a USB webcam to augment the live stream of the camera as described in Section 4.3. The notebook was held by an experimenter or placed on the kitchen counter depending on the preference and position of the participant.

6.3 Methodology

After the participant entered the room and sat down, we discussed the study procedure, study goal, and consent form with the participants, and informed them about their rights regarding data privacy. We collected demographic and medical information before proceeding to the first section of the study.

Section 1, Recognizability. We showed the participant an image of a simple interface consisting only of control elements of the same type. First, no augmentation was present and we asked participants if they recognized the interface element and its function. Then, we showed all seven (or two, respectively) augmentations for that type of interface elements in a fixed order. This order was designed to show similar augmentations, i.e., where only a few aspects change after each other to determine better the impact of single aspects of our design space. Additionally, participants could ask to see previous augmentations again. After each augmentation, we asked them their opinion, specifically if the augmentation makes it harder or easier to perceive the interface elements, how the augmentation could be improved, and which of the augmentations they preferred so far. As the contrast of the augmentation depends on the device’s color, i.e., the background, we created a set of images using different devices. We showed participants these images using their preferred augmentation to investigate the interplay between the augmentation and the background.

Section 2, Real World Usability. In the second section of our study, we wanted to collect feedback about real-world usability, especially when comparing the augmentations to actually touching a real interface. Therefore, we accompanied the participants into our kitchen and asked them to explore the appliances in the room. The kitchen is shown in the top image in Figure 7b and offers two microwave ovens (one with *PushButtons*, one with *TouchButtons*), a stove, a dishwasher, and other smaller devices. After participants reported having fully explored the kitchen and identified the devices and their interfaces, we asked them to briefly explain the layout and

elements of each device. Then, we offered them assistance through our prototype application. Here, we chose the augmentations that the participants preferred in the first section of the study. The participants were handed the camera and shown the live feed on the notebook screen. We first asked them to investigate devices, where they identified the interface elements correctly to let them familiarize themselves with the application. Afterward, we asked them to check the devices where they missed control elements or identified them as the wrong type. We again collected feedback about their opinion, especially regarding the application’s usefulness and if they could recognize more interface elements or identify them more easily.

Section 3, Distinguishability. In the last section of our study, we wanted to evaluate how the augmentations of different types of interface elements are perceived if they appear simultaneously. We showed participants four to nine images of more complex interfaces. In each image, at least two interface elements with at least two different interaction types were visible. Elements of the same type were augmented by the same visualization in each image but with different visualizations in different images. We asked participants about their opinion, especially regarding the interaction of the different augmentations.

6.4 Results

During our study, we collected qualitative feedback about the visual designs of augmentations for interface elements and about the real-world usefulness of our prototype application. We report the participants’ feedback regarding the augmentations’ color and contrast, filling, arrows, and textual information. Further, we summarize the feedback regarding the differences between types of interface elements, the preferred augmentations, and our prototype application. Additionally, we provide a list of the participants’ ratings of the augmentations for *Buttons* and *Knobs* in Table 5 and Table 6.

Color and Contrast. The personal preferences regarding color differed between participants. However, five out of six participants reported green or purple to be hardly perceivable. For all participants, the contrast of an augmentation to its background was more important than the color. On a light gray or white background, six participants could easily perceive the color red, and five participants blue. On the same background, five participants found orange and four participants yellow hard to perceive. However, on

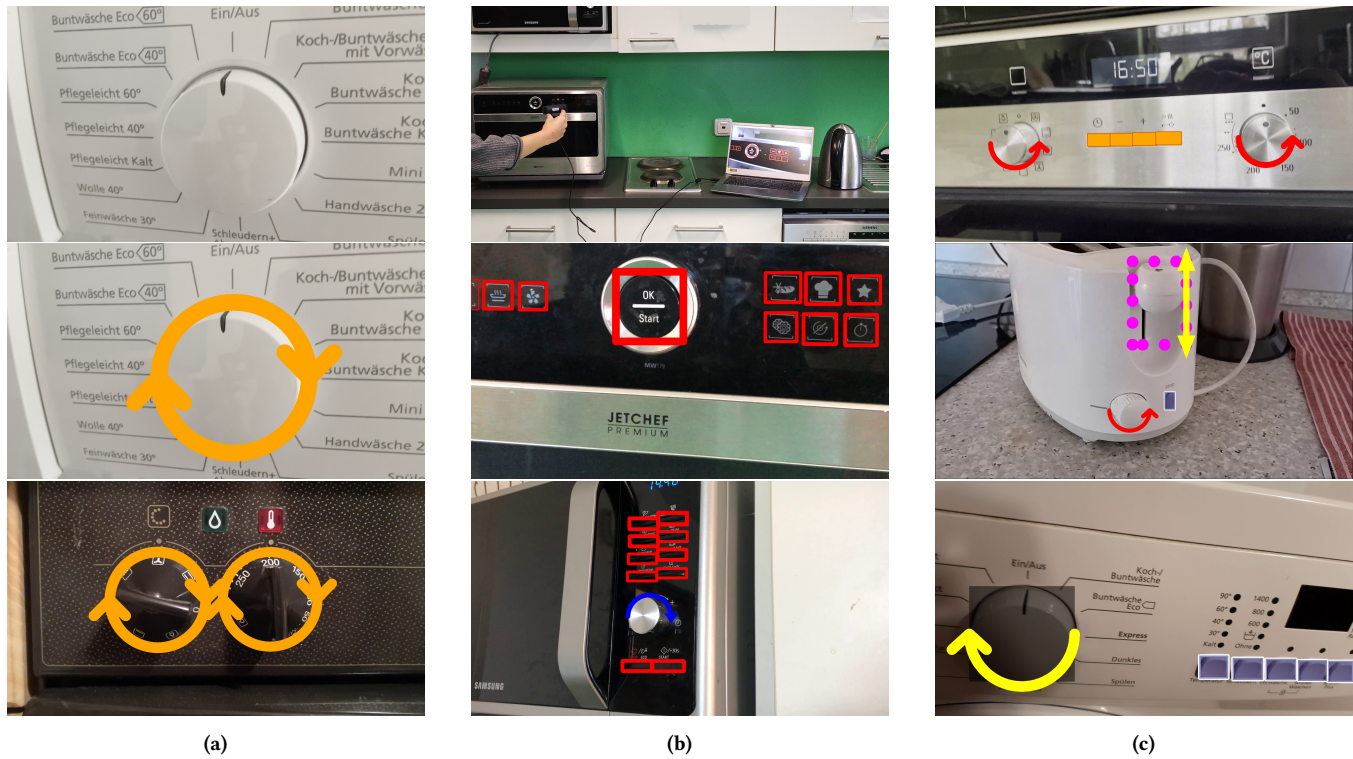


Figure 7: This figure is divided into three sub-figures, each displaying three images. Part (a) shows images used in section one of the user study. The top two images display a picture of a white *Knob* on a white appliance; here, the middle image is augmented. The bottom image shows one of the alternative images with two black *knobs* on a black device. The next part, (b), displays a view of the real kitchen in the top image. A notebook with our prototype application is placed on the kitchen counter and augments the live feed from the connected camera. The following two images show a closer look at the augmented interface of the two microwave ovens. Lastly, part (c) shows three images of the third part of our study. Each image shows a more complex interface, where all interface elements are augmented.

Table 5: This table lists the participants’ ratings of the seven *Button* augmentations from Figure 6. Based on their responses, we assigned a numerical value. Here, 0 stands for “bad”, 1 for “neutral”, 2 for “good”, and 3 for “favorite”.

	P1	P2	P3	P4	P5	P6	Σ
Red Border	2	2	2	3	2	2	13
Red Label	0	1	3	1	0	1	6
Yellow Corners	0	1	0	1	0	1	3
Yellow-Blue Combination	3	3	0	1	3	3	13
Purple Filling	1	1	1	0	0	1	4
Dark-White Combination	1	1	1	0	1	0	4
Opaque Orange	0	0	2	0	2	0	4

darker backgrounds orange and yellow augmentations could be well perceived by three participants. Participant P5 stated, “I would love it if the color was adaptable to the device.” Four participants explicitly asked for maximum contrast, i.e., a black augmentation on white devices, and P1 stated, “The most important is simply the contrast.”

Table 6: This table lists the participants’ ratings of the seven *Knob* augmentations from Figure 6. Based on their responses, we assigned a numerical value. Here, 0 stands for “bad”, 1 for “neutral”, 2 for “good”, and 3 for “favorite”.

	P1	P2	P3	P4	P5	P6	Σ
Red Arrow	2	2	2	2	3	2	13
Blue Arrow	1	3	0	1	2	2	9
Bordered Blue Arrow	0	0	0	0	0	2	2
Yellow Arrow with Filling	3	1	3	1	1	1	10
Double Arrow	1	1	0	0	0	0	2
Bordered Double Arrow	0	0	0	3	0	0	3
Orange Circle	0	2	1	0	1	1	5

Filling. Five out of six participants found the filling of augmentations helpful, as it can be used to create a higher contrast to the border or the arrow, even if the device’s color does not offer such a strong contrast. Further, P1 reported that the gap between the border and filling in *Yellow-Blue Combination* and *Purple Filling* provided a contrast-rich visual stimulus. Other participants were not

disturbed or hindered by this gap but did not find it advantageous. However, participants agreed that the filling should be transparent to not fully obscure the underlying interface element.

Arrows. For the arrowhead, participants did not report a general preference regarding the form and position. However, if the background was not uniform, i.e., one side of the arrow was above a dark background, and one side was above a light background, the position with the higher contrast was preferred. Further, participants inquired about the connection between the arrows pointing direction and the possible turning direction of a *Knob*. Here, five participants would find it intuitive if the arrow not just indicates that a *Knob* can be turned but also aligns with the possible turning directions, i.e., a double-sided arrow is used for *Knobs* that can be turned in both directions. In all cases, three participants highlighted the importance of a wide and not too pointy arrowhead for good recognition of the arrow.

Textual Information. Adding textual labels to augmentations was perceived as redundant due to the already clear interaction type (P1, P4, P6) and annoying and time-consuming to read (P1). Only two participants reported benefiting from the label due to the increased area of the augmentation. Therefore, the same effect could be achieved by a thicker border. However, participants mentioned that a label providing information about the function (and the interaction type) would be more helpful.

Differences Between Interface Elements. Most visual properties were perceived similarly across augmentations and interface elements. However, some aspects were perceived differently for different types of interface elements. Participants P1 and P4 reported that the round arrows in the augmentation for a *Knob* help localize the element, whereas the straight arrows for *Slider* support its intuitive use. Five participants reported that an arrow alone would suffice to augment *Sliders*, and the additional border is "too much" and "annoying". In contrast, four participants reported the additional border is required for *Toggles* since an arrow alone would not be helpful to them.

Prototype Application. When exploring the kitchen, all participants found our application generally helpful. The *TouchButtons* on the microwave oven shown in the middle image in Figure 7b were not visually perceivable by most participants. Often they were only noticed when involuntary inputs were triggered upon scanning the device with their hands. Such an input caused audio feedback, which resulted in participants being surprised and removing their hands from the device. Here, the application could provide the most support, and most participants expressed astonishment about the amount of *TouchButtons* on the device and that the right side has two rows of elements. Similarly, the *Knob* on the same device, which is also a *Button*, was without help often only identified as a *Knob*. For devices that are intuitive to use or only offer a few interface elements, e.g., a kettle, three participants (P1, P3, P4) reported that they do not need such an application daily. However, P3 stated that the application is helpful when interacting with devices where the interface elements have little visual distinction from the device. Further, all six participants found our prototype application helpful in familiarizing themselves with new devices. Participant P2 described

the application by saying, "It worked wonderfully. [...] Especially it would help [...] when you are figuring out a new machine".

6.5 Discussion

Our study shows that visual augmentations are perceivable and helpful. Especially for unfamiliar devices with low-contrast interfaces, participants would use such an application. We found that while individual preferences and perceptions are relevant, high contrast between the augmentation and the device and between different elements of the augmentation is crucial.

High Contrast is Crucial. All participants found high contrast helpful, and five reported it as their primary cue for visual perception. However, some participants reported only the brightness contrast as necessary, while color contrast is not perceived well. We found that contrast needs to be there, but it does not matter how that contrast is created. Therefore, a simple border contrasting the device color or a border with a filling in a contrasting color can both be recognized well. For interface elements with shapes other than rectangular, a rectangular border creates gaps between the border and the element. This proves beneficial, especially if a potential filling can be adjusted to the shape of the interface element.

Individual Perception and Preferences Differ. Even with the same diagnosis and the same visual acuity, the visual perception of people with visual impairment differs. Especially, the color perception varies between participants. Participant P4 stated, "I do not like yellow a lot. [...] If I really need to draw attention to something, then I need it to be red." whereby P5 stated, "I find the version of yellow buttons in contrast to gray very usefully." While some augmentations received high ratings (*Red Border*, *Yellow-Blue Combination*, *Red Arrow*) and some low ratings (*Yellow Corners*, *Bordered Blue Arrow*, *Double Arrow*) by the majority of the participants, individualization remains a key factor for assistive technology. However, the visually simple and high-contrast design of the good-rated augmentations does provide some insights into augmentations that most people with visual impairment can perceive.

Context Aware Augmentations are More Intuitive. We already discussed that augmentation should provide high contrast to the device they are augmenting. However, other adaptations to the real world are also perceived as helpful. The augmentation should not extend over the edge of the device and consider the interaction possibilities of the augmented interface element. For example, the arrow on a *Knob* should indicate all possible turning directions.

Augmented Reality Applications Support Users. While not all participants would always use such an application, it can support users in familiarizing themselves with new devices. It especially provides a benefit for household appliances with many interface elements that do not have high contrast to the device itself or offer no tactile feedback. Even perceiving a schematic of the interface layout without information about the function of individual elements provides a benefit to the user. Our study shows that even a prototype application implemented with a notebook and a webcam is perceived as helpful and that people with visual impairment desire such assistive technology. Future applications can be implemented on mobile devices to increase usability and integrate function recognition to increase the application's usefulness.

7 LIMITATIONS AND FUTURE WORK

The presented dataset and prototype application have some limitations. In this section, we discuss these limitations and other points with potential for improvement, so that future work can build on this research when designing similar applications.

Dataset. In modern appliances, *Slider* and *Toggle* have become uncommon interface elements. Hence, these two classes are under-represented in the dataset. Additionally, these interface elements are not uniform in their design, with the result that the neuronal network did not recognize these classes as well as the others. On Github [39, 40], we provide instructions on how others can extend the dataset. Such contributions may help in creating a more versatile neuronal network.

Evaluation. To evaluate our prototype application, we recruited six individuals with diverse residual vision capabilities. We conducted a qualitative study to investigate the general usability of a system such as ours. Future evaluations will include more participants and a focus on a quantitative study. A higher number of participants will also increase the diversity in terms of the participants' residual vision capabilities and the generalizability of the results.

Prototype Application. In our prototype application and study, we chose to use a screen-based approach. The next step towards an application that can support people with visual impairment in their everyday life is integrating such a system into mobile devices like smartphones or head-mounted displays (HMDs). While smartphones are already widely distributed, they offer only a small screen and occupy one hand during use. HMDs, such as the Microsoft HoloLens, are improving fast and are likely going to be used by the general public within the next decades. Research on the use of HMDs as assistive technology is widespread [23, 24, 47, 48] and HMDs for people with visual impairment already exist as off-the-shelf devices [7, 31]. Future research should investigate the usage of HMDs for augmenting the interface of household appliances while considering computational resources and data privacy in case an image is sent to a server for recognition. However, in see-through devices, developers must consider that black or dark colors appear transparent, while other colors provide additional contrast due to the increased brightness in relation to the environment. The augmentations we proposed in this work have to be adjusted accordingly.

Augmentations. To improve the contrast of the displayed augmentations, the background colors in the image could be recognized and the augmentation adjusted accordingly. For example, one can calculate the average color of all pixels inside a detected bounding box and choose the complementary color for the augmentation. Alternatively, more sophisticated approaches can maximize the contrast by considering more details of the background [46]. The visual output can also be supplemented by audio or tactile feedback. Further, the system can be extended to recognize the function of different interface elements, the current state of the appliance as well as the user's intention to provide support specific to the situation at hand.

8 CONCLUSION

We created a dataset containing 13,702 labeled images of interfaces from household appliances and show that this dataset can be used to successfully train a neural network to identify the location, size, and type of interaction of interface elements.

We designed 18 different visual augmentations for interface elements based on our design space for the four types of interface elements, namely *Button*, *Knob*, *Slider*, and *Toggle*. Afterward, we conducted a user study to evaluate the augmentations as well as a prototype system showcasing functionality using a real kitchen. Our prototype was well received by all participants, especially regarding the recognition of *TouchButtons*.

We show that our dataset is useful for training a system recognizing interface elements and that visual augmentations for interface elements are recognizable and helpful. Such an application aids people with visual impairment in learning how to interact with unfamiliar devices and supports them through interactions with familiar devices. Here, the automatic parsing and augmentation of interfaces provide an important step toward the independent interaction of people with visual impairments with their everyday environment.

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